Tangents to a circle

1 The diagram shows an arbitrary point C on the $\overleftrightarrow{AD}$, and B is the point at which the tangent meets the circle.



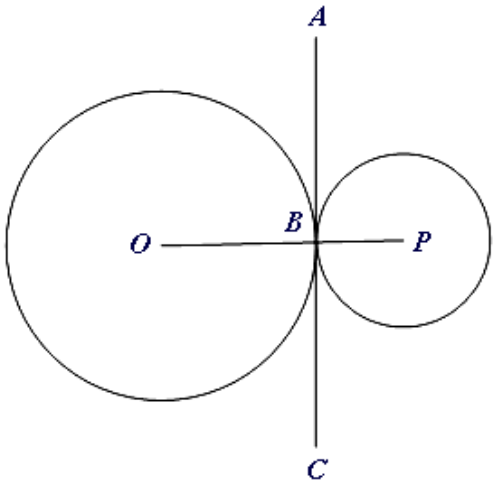
a Is $\overline{OB}$ greater than, less than, or equal to $\overline{OC}$?

b What point on $\overleftrightarrow{AD}$ is closest to the center of the circle?

c In general, what can we say about the angle of a line joining some point to some other line by the shortest route?

d Hence, what can we say about $\angle OBA$?

2 Consider the following diagram:



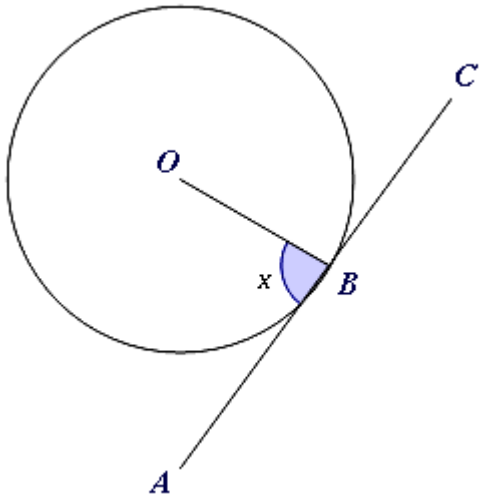
a Show that $\angle OBP$ is a straight angle.

b Hence, what can we say about points O, B and P?

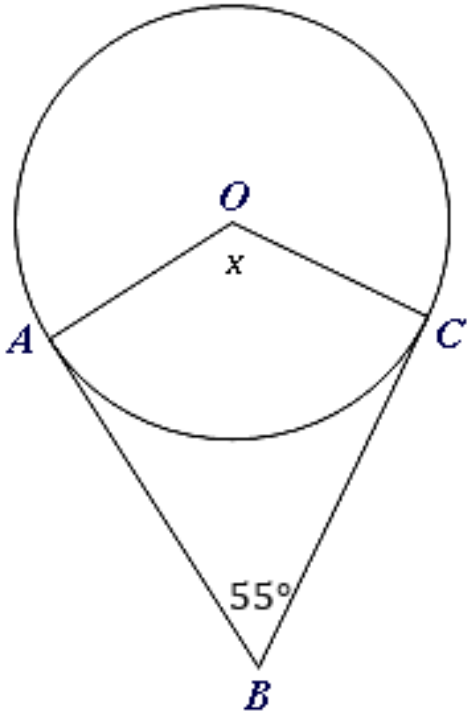
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3 Find the measure of the variables for the given diagrams. Explain your reasoning.

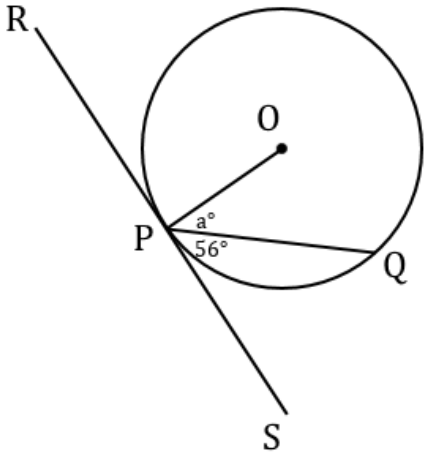
a



b



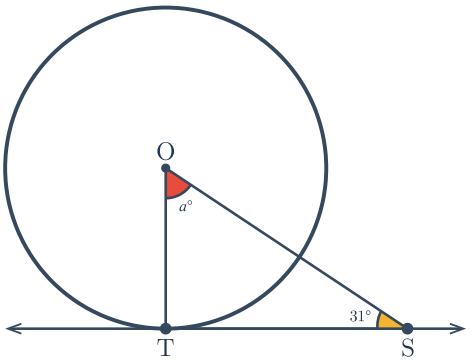
c



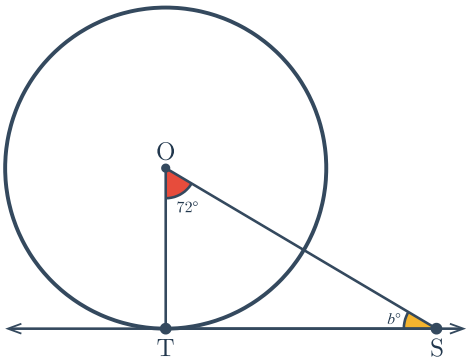
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4 Find the value of the variables for the following diagrams:

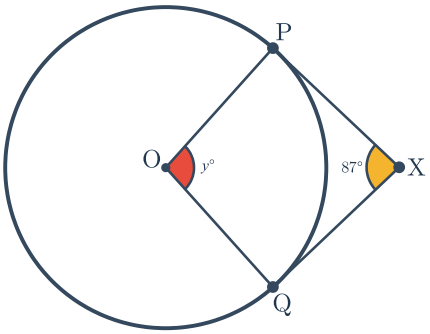
a



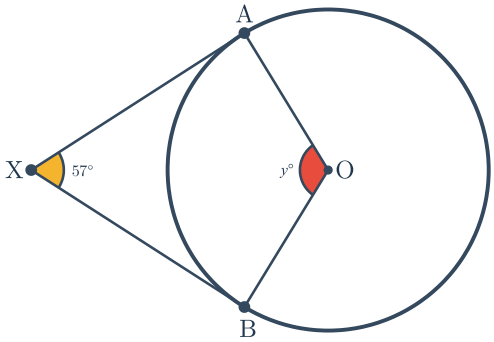
b



c



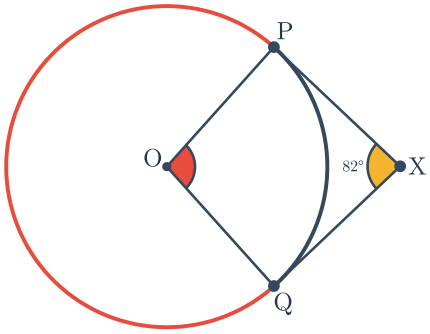
d



5 The diagram shows $\overline{PX}$ and $\overline{QX}$ as tangents to the circle at P and Q, respectively.

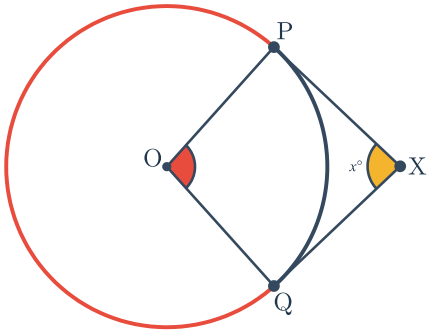
a Find the measure of the central angle subtended by $\overset{\frown}{PQ}$.

b Find the measure of the major intercepted $\overset{\frown}{PQ}$.



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6 The diagram shows $\overline{PX}$ and $\overline{QX}$ as tangents to the circle at P and Q, respectively.



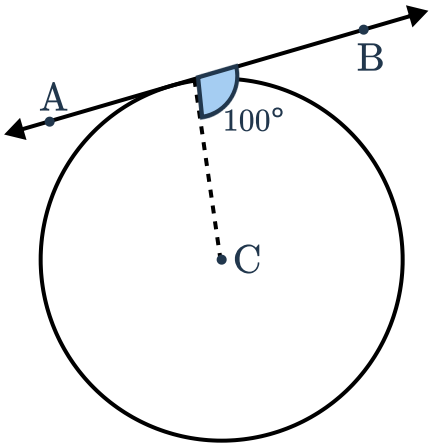
a Find the measure of the central angle subtended by $\overset{\frown}{PQ}$ in terms of x.

b Find the measure of the major intercepted $\overset{\frown}{PQ}$ in terms of x.

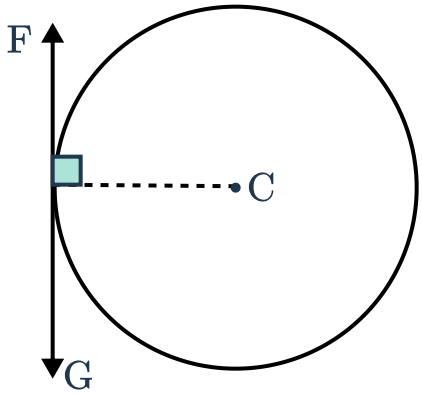
Additional questions

7 Determine whether each line is tangent to the given circle:

a

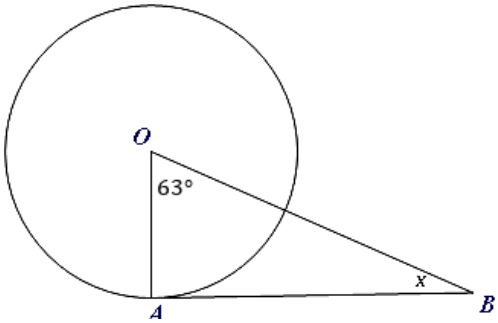


b



8 $\overleftrightarrow{BC}$ touches circle A at point C. The shortest distance from $\overleftrightarrow{BC}$ to the center of circle A is equal to the radius. Is $\overleftrightarrow{BC}$ a tangent line to circle A?

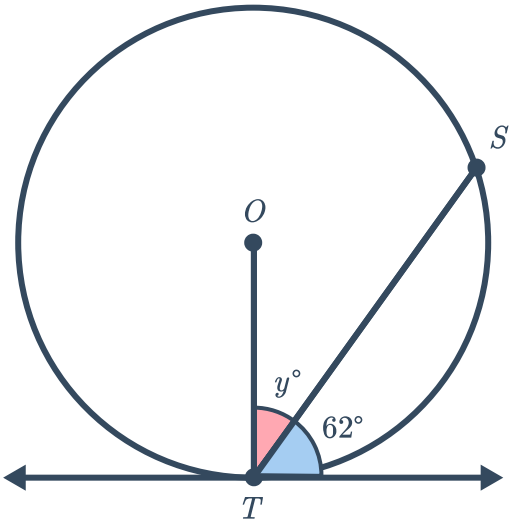
9 If $\overline{AB}$ is tangent to circle O, what is the value of x?



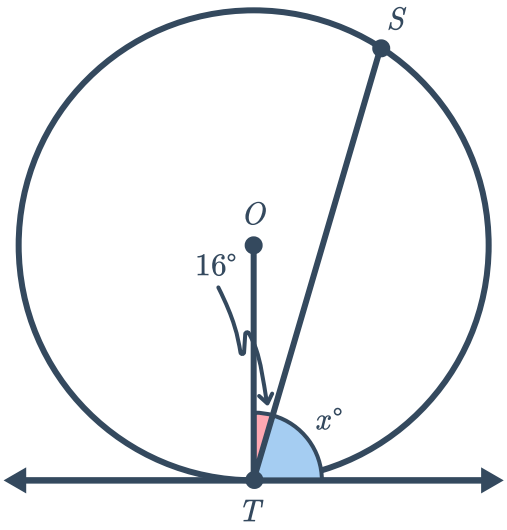
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10 In each diagram, the line through T is tangent to the circle O.

a Solve for y.

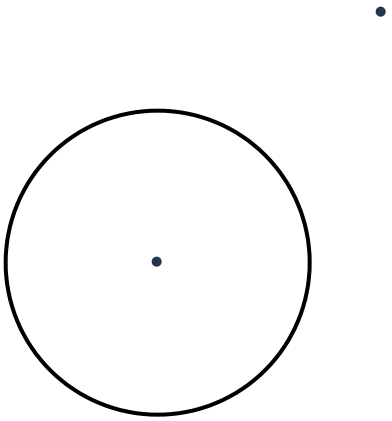


b Solve for x.

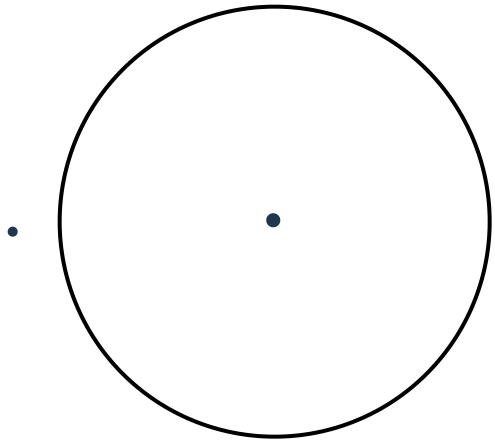


11 For each circles, construct all lines that are tangent to the circle and pass through the given point:

a



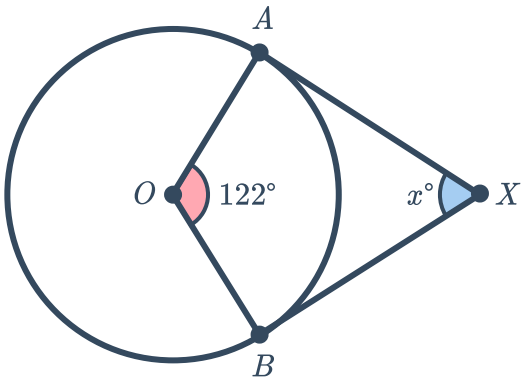
b



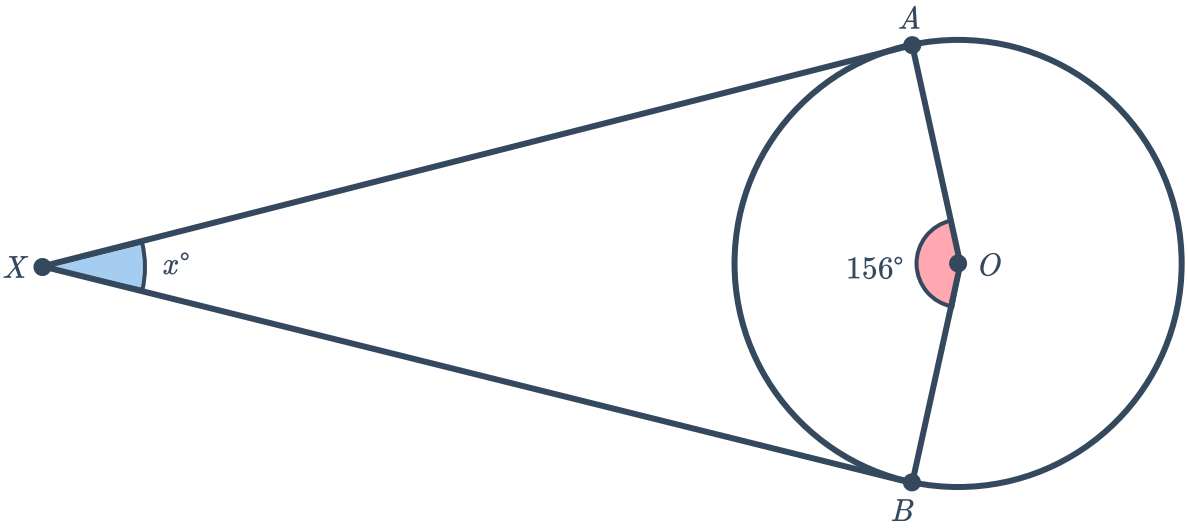
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12 In each diagram, $\overline{AX}$ and $\overline{BX}$ are tangents to the circle O at A and B respectively. Solve for x.

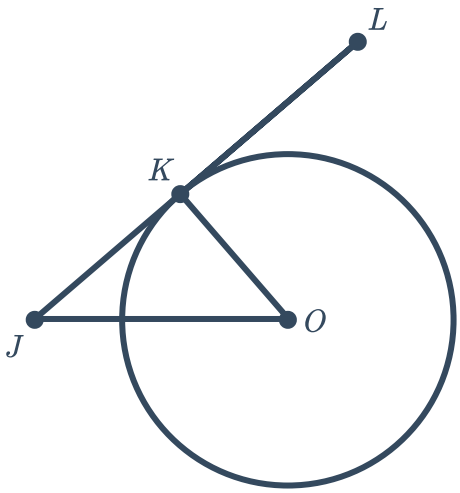
a



b

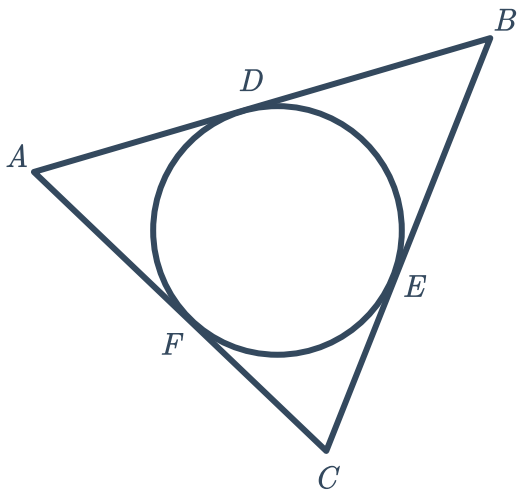


13 $\overline{JL}$ is a tangent to circle O. $JO = 29$ and $JK = 20$.
Solve for OK.

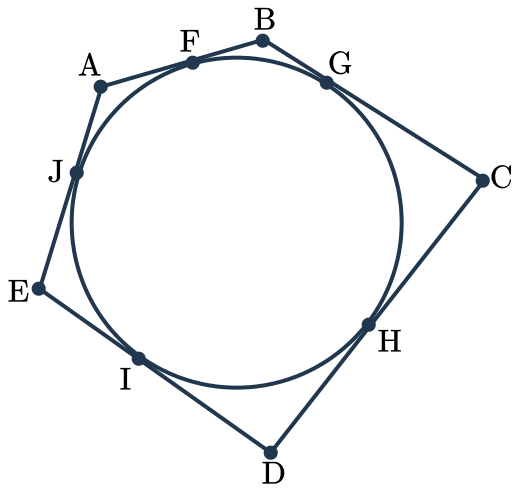


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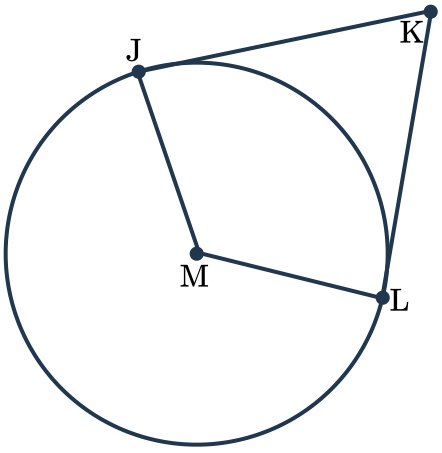
14 $\overline{AB}$, $\overline{BC}$, and $\overline{AC}$ are tangents to the circle. $BD = 7$ cm, $AF = 6$ cm, and $BC = 11$ cm. Find the perimeter of $\triangle ABC$.



15 $\overline{AB}$, $\overline{BC}$, $\overline{CD}$, $\overline{DE}$ and $\overline{EA}$ are tangents to the circle. $AB = 14$, $BG = 5$, $GC = 16$, $DE = 9$ and $EJ = 3$. Find the perimeter of $ABCDE$.

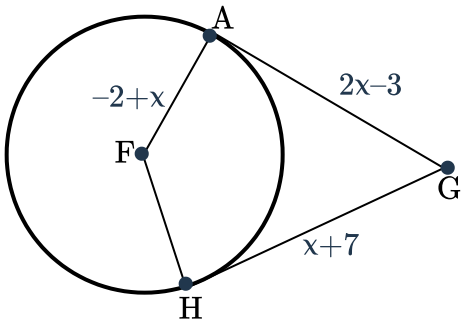


16 $\overline{JK}$ and $\overline{KL}$ are tangents to circle M. $JK = 12x - 4$, $JM = 5x - 12$ and $ML = 2x + 6$. Solve for KL.

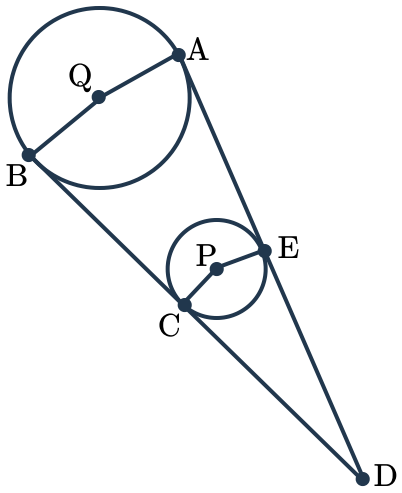


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17 $\overline{AG}$ and $\overline{GH}$ are tangents to circle F.
Solve for FH.



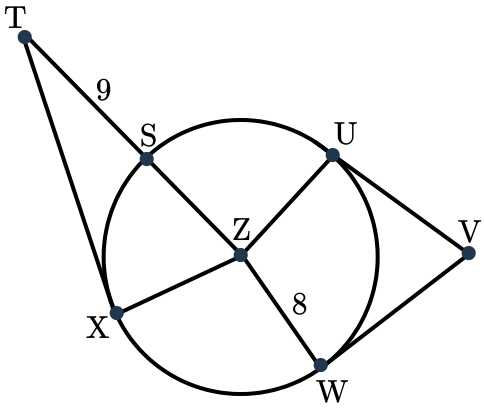
18 $\overline{AD}$ and $\overline{BD}$ are tangents to both circle Q and circle P. $AE = 14$, $PC = 5$, and $MD = 8$.
Solve for BD.



19 $\overline{XT}$, $\overline{UV}$ and $\overline{WV}$ are tangents.
$m\angle UZW = 2x + 15$ and
$m\angle UVW = 6x + 13$.

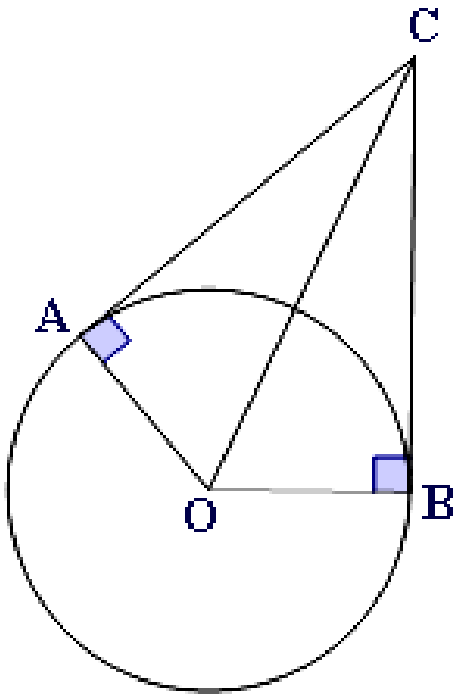
a Find x.

b Find XT.



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20 Given: $\overline{AC}$ and $\overline{CB}$ are tangent to circle O
Prove: $\overline{AC} \cong \overline{BC}$

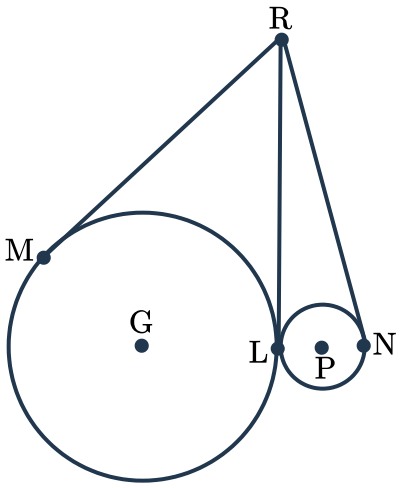


21 When a scoop of ice cream is put into an ice cream cone it falls until it gets to the point in the cone where the scoop is too big to fall any further.
Assuming the scoop of ice cream is spherical, describe the mathematical conditions which must be true for half of the scoop to fit inside the cone.

22 Given:

- $\overline{RM}$ is tangent to circle G
- $\overline{RL}$ is tangent to circle G and circle P
- $\overline{RN}$ is tangent to circle P
- circle G is not congruent to circle P

Explain why $\overline{RM} \cong \overline{RN}$.



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